

River Deep Mountain AI

Model Benchmarking Report: Bare Cropland (BB2Aii)



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FINAL REPORT

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Glossary

BSI	Bare Soil index
CROME	Crop Map of England
GEE	Google Earth Engine
NDMI	Normalized Difference Moisture Index
NDSI	Normalized Difference Snow index
NDVI	Normalized Difference Vegetation Index
OSM	Open Street Map
RDMAI	River Deep Mountain AI
SAVI	Soil Adjusted Vegetation Index
WFD	Water Framework Directive

Summary

Table 1-1 Benchmarking summary

Model Outputs	Inference from CROME data	RDMAI Model
Model Approach	CROME data is an annual land use identification dataset derived from satellite imagery. Given the land use, it is possible to infer sowing date and thus the state of ground in the preceding winter.	All processing is carried out within the Google Earth Engine (GEE) Code Editor environment. The user needs to define the study area and create a crop mask (delineating the arable area) from available data sets and upload this into GEE. The satellite imagery required by the model for the study area is retrieved from within GEE. The model then does all the processing (masking cloudy pixels, generating median seasonal composites, and calculating vegetation and soil indices) and predicts bare crop land during the time frame specified by the user.
Applications	CROME data is widely used in modelling tasks and spatial analysis where identification of land cover is required.	Identification of the duration and extent of bare cropland. This could be used to inform decisions about where to target pollution control measures, such as the use of cover crops.
Geographic Constraints	CROME data is available for England. Alternative data sources with wider geographic coverage are available (potentially with cost implications)	The model is not inherently restricted to the UK, but the model makes use of crop masks that are, in order to filter the results.
Temporal Resolution	CROME data is produced annually.	The script can be modified to use any date range depending on user requirements.

Data Inputs Required	CROME data and an expert-judgement based list of which CROME crop categories would be preceded by bare cropland over winter.	The required satellite data is automatically retrieved from GEE.
Model Performance	Without an observed dataset, it is not possible to provide a quantitative assessment of accuracy.	Without an observed dataset, it is not possible to provide a quantitative assessment of accuracy. However, the RDMAI model achieved an accuracy of 70% if the CROME data is taken as being correct.
Cost & Time	CROME data is freely available. Once the data for the area of interest is downloaded, producing the bare cropland output is a straightforward GIS task.	Application of the model took a few hours, but could be more efficient depending upon the task and user familiarity with the model. The only potential costs are for GEE, but the costs for the processing undertaken in this benchmarking were negligible.
Strengths	Simple. Freely available dataset, updated annually.	Relatively simple once user is familiar with the model. Should capture any over-winter management, such as cover crops, or issues such as winter crops not establishing well. Can produce a time series, showing how long the land is bare over the winter.
Limitations	Not all CROME land use categories allow an estimation of sowing date (e.g. mixed cropping categories, or where a specific crop can be either spring or winter sown). Even though a crop is sown in the spring, that does not mean that the land was bare in the preceding winter.	Availability of cloud-free imagery Requires a GEE account

	Predictions are for the winter preceding the CROME data, so cannot predict winter cover in the current winter period, thus more suited for longer-term strategic use than for near-real time activity.	
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1. Benchmarking Methodology

1.1 Model Selection

1.1.1 RDMAI Bare Cropland Model Overview

The Open Bare Cropland Detection Model identifies crop fields left bare or ‘fallow’. Bare fields are at higher risk of erosion and nutrient runoff and are a potential source of agricultural diffuse pollution. Using Sentinel 2 imagery and spectral indices such as Normalized Difference Vegetation Index (NDVI), Soil Adjusted Vegetation Index (SAVI), Bare Soil index (BSI), Normalized Difference Moisture Index (NDMI), and Normalized Difference Snow index (NDSI) within Google Earth Engine (GEE), the model identifies seasonal soil exposure and generates time series outputs to support evidence-based catchment management.

The model is intended to allow stakeholders to understand how areas of bare cropland change within a catchment at sub-annual timescales, enabling more effective targeting of catchment interventions that can reduce the risk of soil erosion and nutrient runoff.

1.1.2 Overview of Existing Industry Model(s)/Tool(s)

There is no existing model or dataset that shows the locations of bare cropland. However, there are existing datasets showing land use and cropping, from which it is possible to infer the state of ground in the previous winter. The two key such datasets for this are the UKCEH Landcover maps plus crops dataset ([UKCEH Land Cover plus: Crops](#)) and the Crop Map of England ([CROME](#)). The Landcover dataset covers the whole of the UK, but requires a paid licence for use, whilst the CROME data is free to use, but for England only. Given that the RDMAI model is freely available, the CROME data has been used for this benchmarking.

1.2 Catchment Selection

For the benchmarking, 10 catchments were identified that should cover a range of agricultural and environmental conditions across England. The process for catchment selection, based on the c. 4,000 WFD waterbodies in England, was

1. Restrict to catchments of comparable size (c. 25km²)
2. Restrict to catchments with at least 40% farmland according to OSM data
3. Restrict to one catchment per 50 x 50 km square
4. Select 10 catchments at random, ensuring there is a range in terms of the proportion of arable land use as a fraction of total land use.

The 3rd criteria listed above was designed to provide contrasting rainfall and soils between the catchments. The 4th criteria was to provide contrasting farm management between the catchments (highly arable catchments with lots of cereals, more grassland catchments where maize and fodder might be more common). The data for criteria 4 was drawn from the cropping information contained within the Farmscoper model.

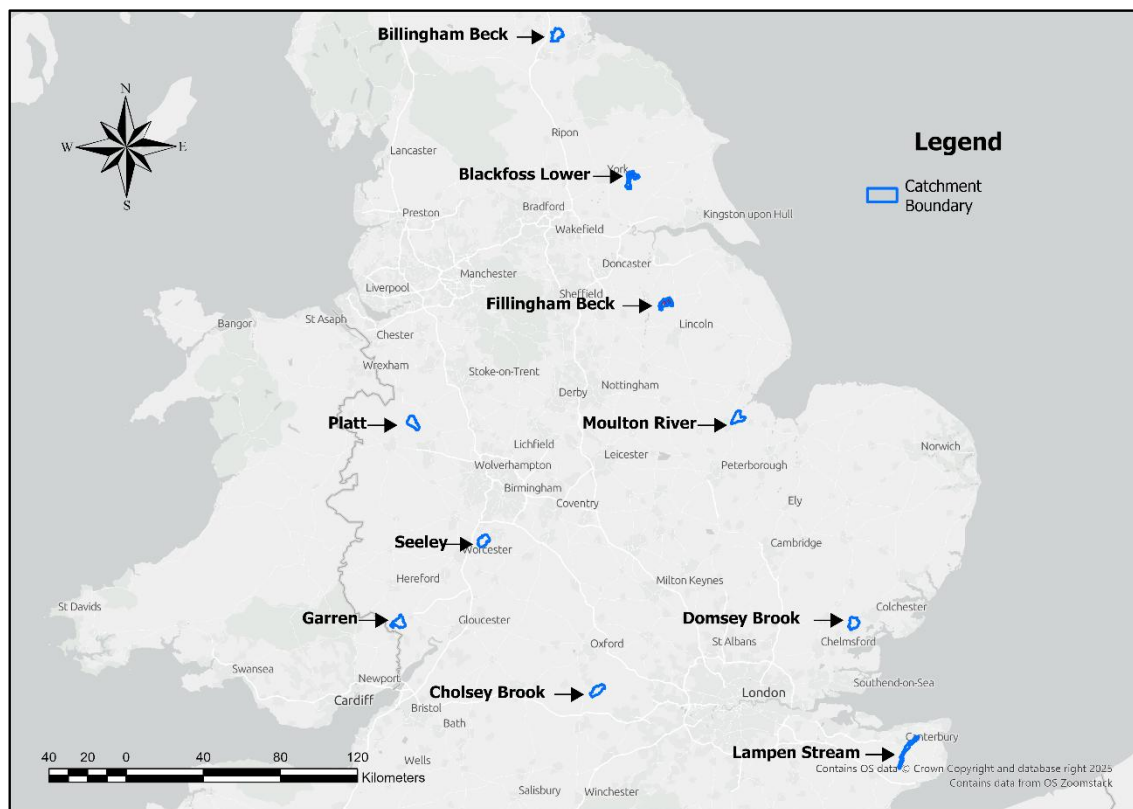


Figure 1 The 10 WFD waterbodies used for benchmarking the bare cropland model.

1.3 Benchmarking Methodology

The bare cropland model was run for all of the catchments and used to locate bare land between December 2023 and February 2024. Aside from changing these dates, there were no other significant modifications to the code taken from GitHub. As satellite data is automatically retrieved from GEE there is no need to purchase or prepare any additional datasets.

The CROME data was obtained for 2024, and an assumption made for each crop category on whether the land would have been bare in the winter of 2023-24 (yes for spring sown crops, no for winter sown crops). Some of the CROME categories do not allow an estimate of sowing date, either because the category is ambiguous, or the crop can be sown in autumn or spring (and is not explicitly labelled as such, e.g. spring barley). These were labelled as unknown. Unknown data occupied between 2 and 10% of the agricultural land – for analysis, this was treated as non-bare land.

The CROME data is provided as hexagonal cells, with each cell 0.41 ha. The RDMAI model works on field parcels, and there would typically be multiple CROME cells within a field parcel. The statistics in this report were calculated using the individual CROME cells, rather than determining the dominant result in a field parcel based upon the data for all the CROME cells within it.

1.4 Benchmarking Limitations

The Cropland detection model is able to identify the temporal extent of bare soil, whereas the inference from the CROME data is the expected management before the CROME crop is sown, thus the full utility of the Cropland model is not being assessed.

The CROME-based approach will not be able to determine if there was over-winter cover due to other factors, for example, a cover crop or rotational grassland ploughed out in spring, and so it is assumed that the CROME-based approach should over-predict the extent of bare cropland compared to the RDMAI model, which does not have this limitation.

CROME data is based on satellite imagery and is thus not necessarily correct. As mentioned in Section 1.3, it is not always possible to infer sowing date from CROME categories.

2. Benchmarking Results

2.1 Model 'Ease-of-Use'

Table 2-1 Comparison of model 'ease-of-use'

Model Outputs	Inference from CROME data	RDMAI Model
Installation & Run Process	An R script was coded that used a CROME LUCODE lookup table to identify land-cover types classified as bare cropland (via Yes/No assignments to individual LUCODEs). The script read the lookup table and a geodatabase containing CROME data clipped to ten catchments, automatically identified layers containing a LUCODE field, and extracted polygons matching the bare-cropland classification.	The installation and run process for the model were relatively straightforward. The documentation clearly states that a Google Earth Engine (GEE) account is required, and all processing is carried out within the GEE Code Editor environment. However, users do need a basic working knowledge of GEE, particularly the asset upload process, how to correctly reference asset paths within scripts, and general navigation of the interface. An OSM layer was intersected with a CROME land-cover dataset to generate a refined agricultural mask. The resulting shapefile was uploaded as a GEE asset and used by the script during model execution. Aside from this intersected mask, the only other essential input are the study area boundaries, also retrieved from this GEE asset. After the required asset is prepared, the user can copy the provided 'gee_detect_bareland.js' script into the GEE Code Editor, update the asset location, and run the model. Some basic knowledge of .js is useful but not necessary.

		Overall set up and running (once understanding of the methodology) takes around 1 hour: 40 minutes downloading, editing and uploading of assets; 10 minutes of script editing and running; 10 minutes exporting and importing outputs.
Data inputs	CROME data clipped to the 10 catchments, and a table based on expert judgment listing which CROME land use codes would be spring sown (i.e. have bare cropland over the preceding winter).	The required inputs for the model are the study area boundaries, defined as polygons of the 10 target catchments. Satellite data is automatically retrieved from GEE (Sentinel-2 surface reflectance), so no manual download is necessary. Data inputs include a filtered OSM farmland layer, which can be used to highlight agricultural areas, and crop type data from the CROME dataset. These datasets were intersected and then clipped to the 10 target catchments. The resulting shapefile is uploaded as a single GEE asset. Most pre-processing steps are handled automatically by the script. This includes masking cloudy pixels, generating median seasonal composites, and calculating vegetation and soil indices (BSI, NDVI, SAVI). Users do not need to perform these steps manually, making the workflow largely automated once the study area is defined.
Licensing Requirements	None for CROME data.	A GEE account is required to run the model. GEE operates on a “pay-as-you-go” system, where users are charged only for the computational credits used. For small study areas, this is very cost-effective. During testing, GEE provided an initial free credit allocation of \$250. Running the model for the catchments used a negligible amount

		of credits, less than \$0.01 per run. No additional licenses or paid software are required.
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2.2 Model Outputs

Table 2-2 Comparison of model outputs

Model Outputs	Inference from CROME data	RDMAI Model
Data Format	CROME data and the output of the model were shapefiles.	The model output is in GeoJSON format, which can then easily be converted into a shapefile.
Temporal Resolution	CROME data is produced annually.	The model analyses imagery over a specified time period. The script was configured to generate composites for winter (December–February) and summer (June–September), although only the winter output was utilised in the benchmarking. The temporal resolution could be modified to use other date ranges depending on user requirements.

Geographic Constraints	CROME data is only available for England. The same approach could have been undertaken using the (paid for) UKCEH Land Cover Plus Crops dataset, which has UK coverage.	The model can be applied to any geographic region covered by Sentinel-2 imagery in GEE. In practice, the original documentation focuses on UK catchments. Optional input layers such as OSM farmland or CROME crop-type datasets are UK or England (For CROME) specific, so their use is limited outside the UK. The model itself is not inherently restricted to the UK, but accuracy may be affected where auxiliary data (e.g., crop masks) are unavailable or incomplete.
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2.3 Time and Cost Comparison

The total runtime for the applying the RDMAI model to the 10 catchments, excluding asset preparation and uploading, was approximately three hours. If all catchments had been pre-processed and combined into a single shapefile, then uploaded as a single GEE asset, the model would only have needed to be executed once, improving efficiency and cost effectiveness.

The CROME dataset is large, containing 32 million hexagonal polygons for the whole of England, and so can take a while to download. The large file size meant it was easier to extract the data for processing for each catchment from county-level CROME data and then combine afterwards. However, once the data was ready, producing the outputs was quick, requiring a simple table to shapefile join based on the CROME crop codes.

Some of these time issues for both approaches are related to the nationally distributed nature of the benchmarking task, and would likely not be encountered by a more typical user focussed on a single smaller area.

2.4 Model Performance

Comparisons of the CROME based and RDMAI predictions for the 10 catchments are shown in Figure 2 to Figure 11. There is often a good visual agreement between the datasets, although for the Platt (Figure 6), the RDMAI model has most of the interior of the waterbody as bare land, whereas the CROME data identifies more bare land on the edges, likewise for the Domsey Brook (Figure 11).

If the CROME approach is assumed to be correct, then it is possible to produce a confusion matrix and associated accuracy calculations (Table 2-3). The overall accuracy, or agreement between the two datasets, is just under 0.70, ranging from 0.59 for the Moulton waterbody to 0.78 for Seeley.

CROME predicting bare land but the RDMAI model predicting cropped land is less common than RDMAI model predicting bare land and CROME predicting cropping. This is as anticipated, as the CROME data is an estimate based purely on sowing date with no knowledge of the actual overwinter management – the use of cover crops or not ploughing in grassland until just before a spring sown crop are two reasons for not having bare cropland that would not be known based on CROME data but should be detected in the RDMAI model.

There is a moderate correlation between the accuracy and the percentage of arable land in the catchment (accuracy is lower when there is more arable land, Domsey, Fillingham and Moulton). However, this correlation may be due to other environmental factors (e.g. rainfall or soil trafficability) and given the relatively limited dataset has not been further investigated.

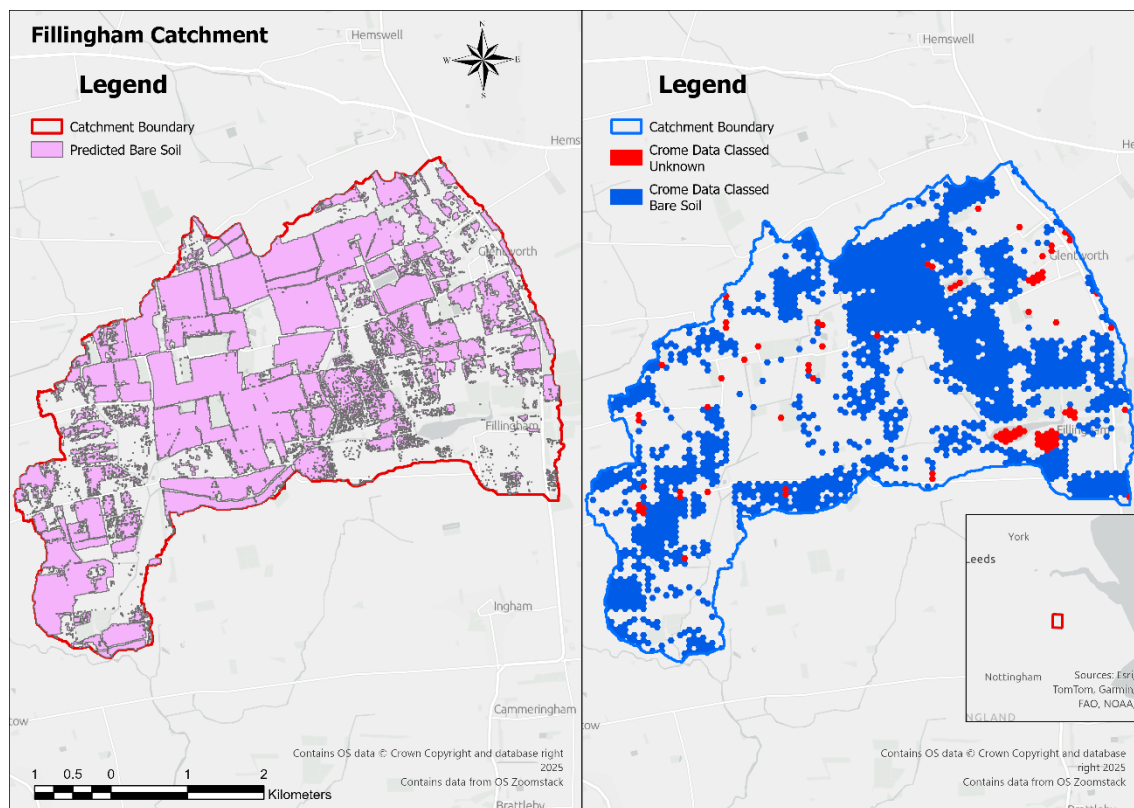


Figure 2 Bare cropland data for the Fillingham Beck WFD waterbody (GB105030062490)

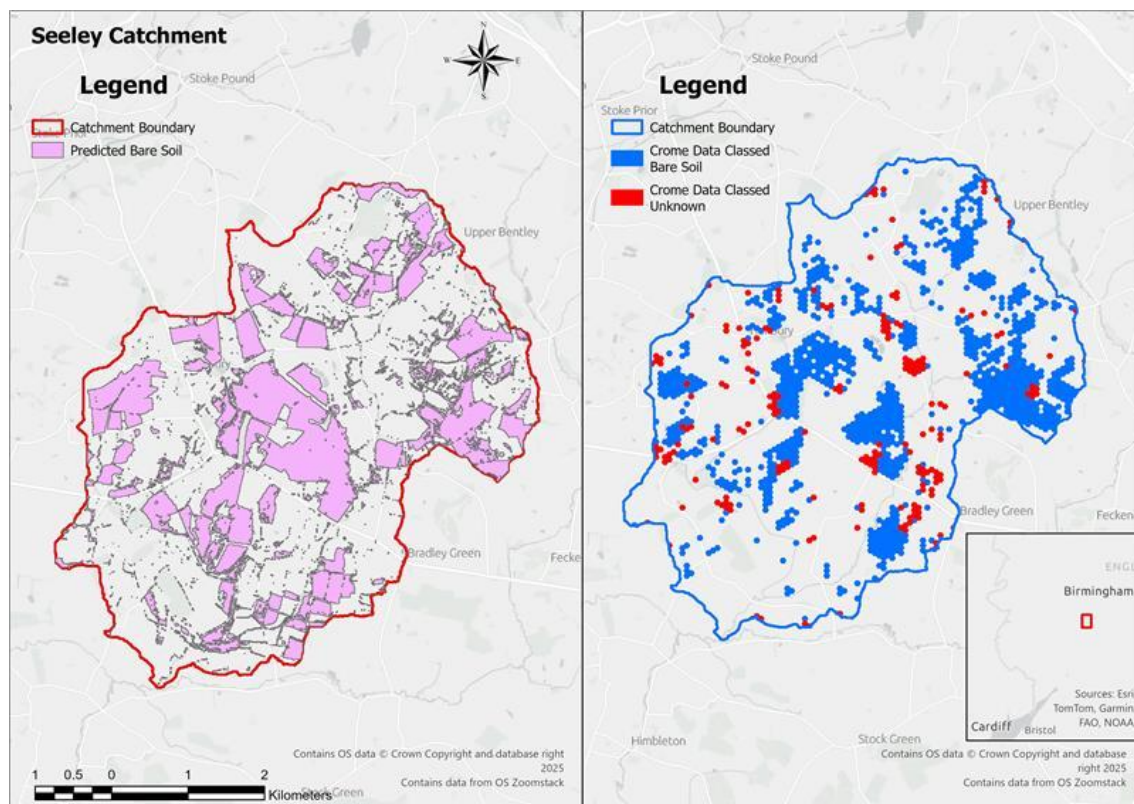


Figure 3 Bare cropland data for the Seeley Bk WFD waterbody (GB109054043790)

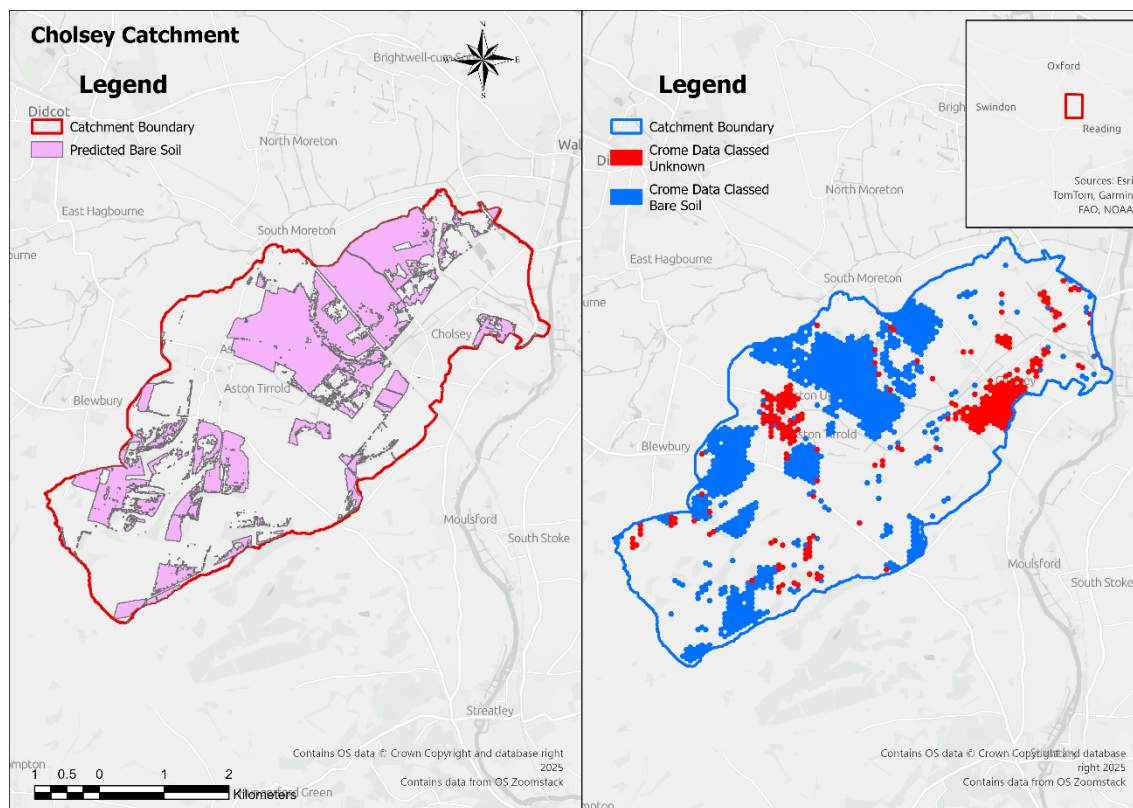


Figure 4 Bare cropland data for the Cholsey Brook WFD waterbody (GB106039023570)

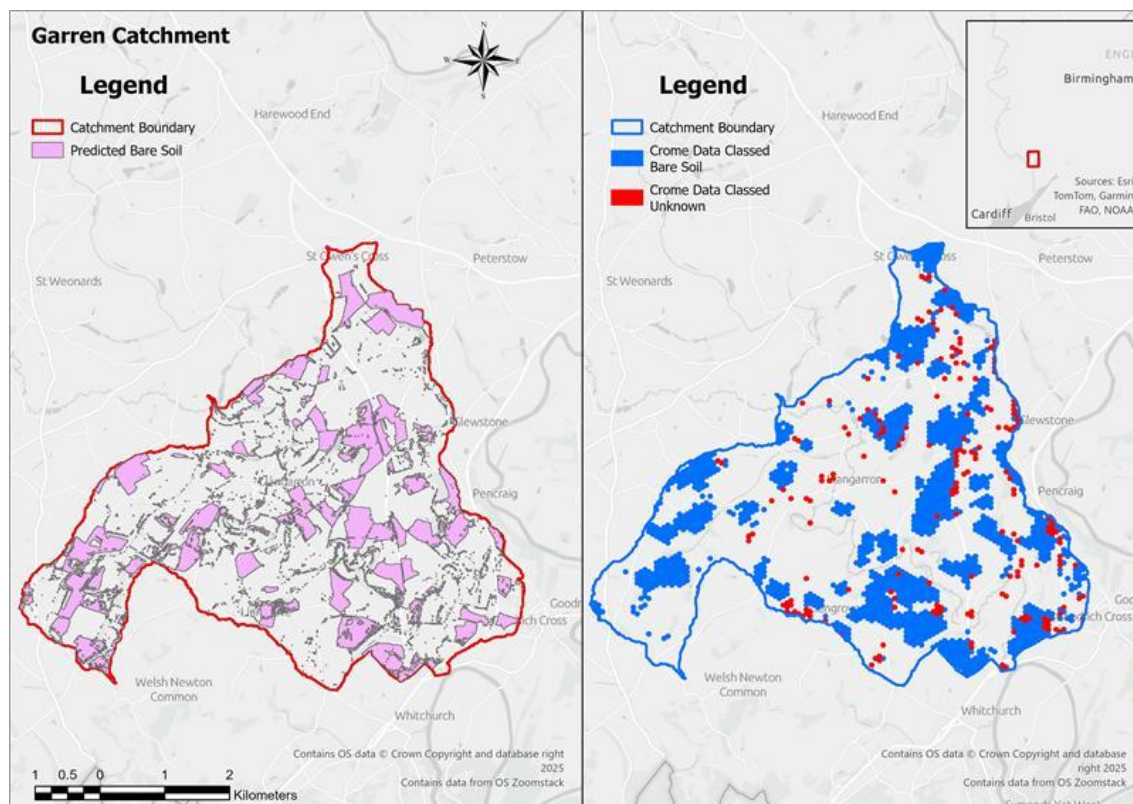


Figure 5 Bare cropland data for the Garren Bk WFD waterbody (GB109055029690)

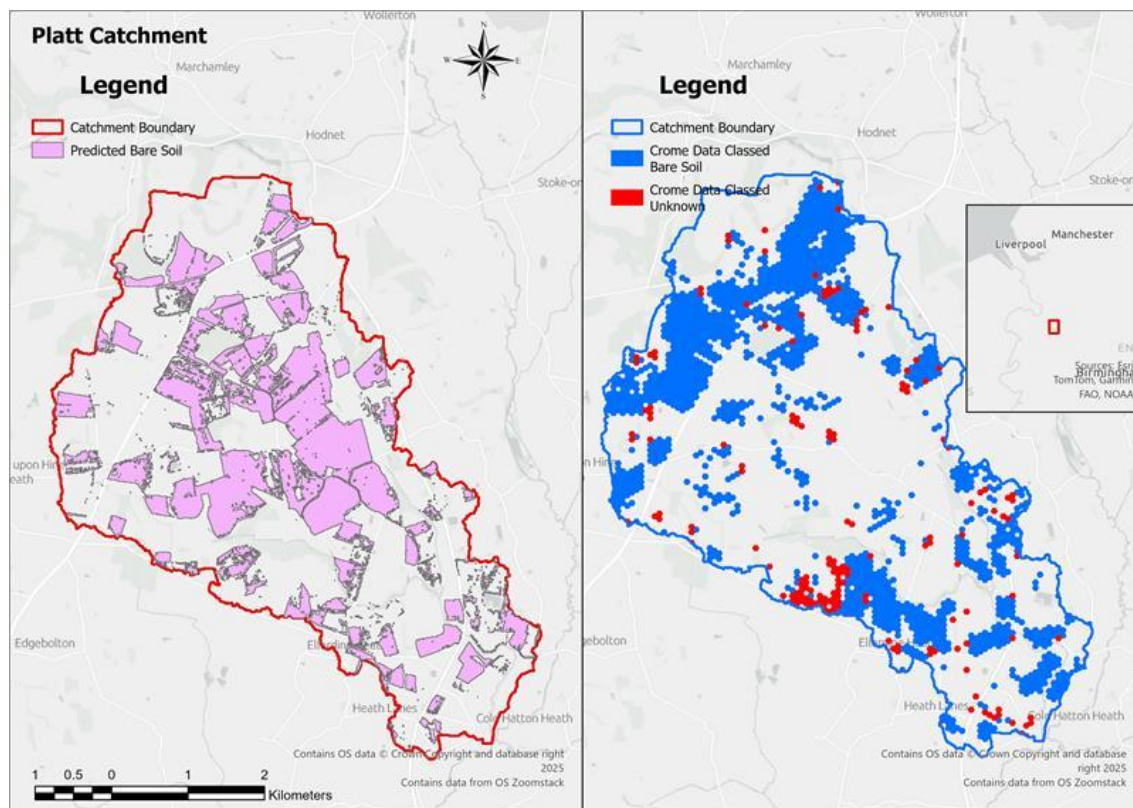


Figure 6 Bare cropland data for the Platt Bk WFD waterbody (GB109054050210)

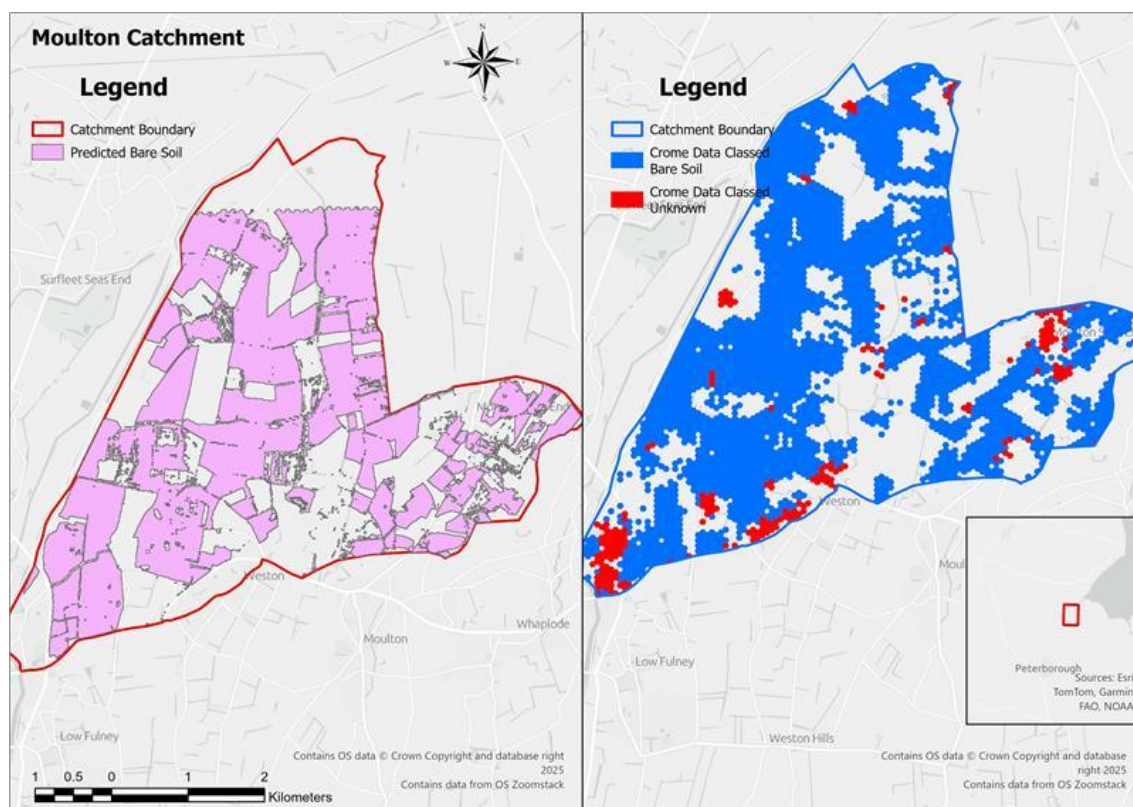


Figure 7 Bare cropland data for the Moulton River WFD waterbody (GB205031050755)

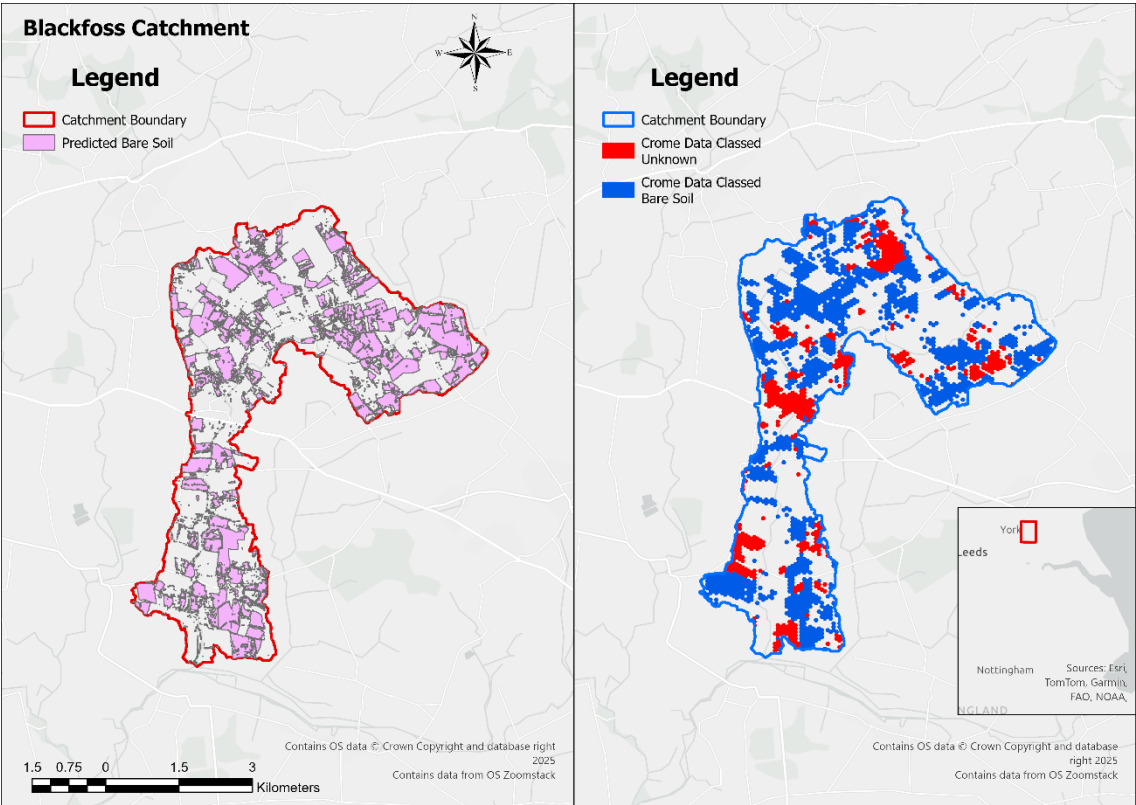


Figure 8 Bare cropland data for the Blackfoss Bk WFD waterbody (GB104027063460)

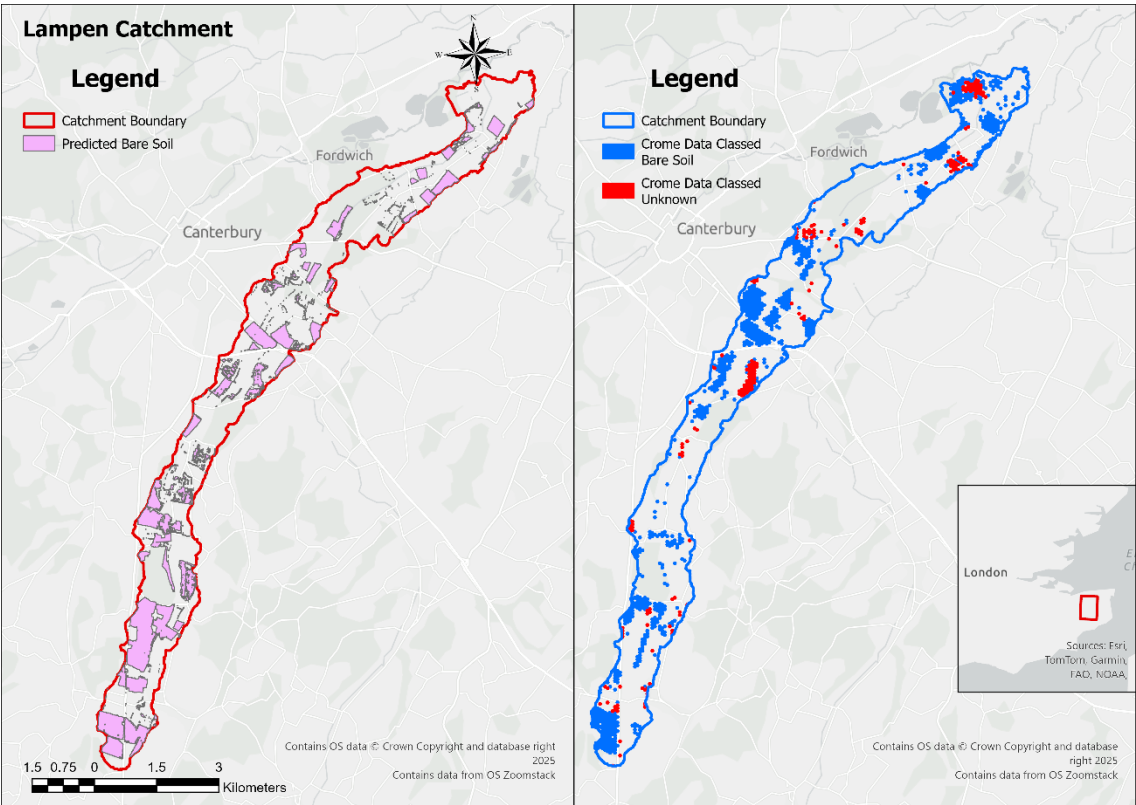


Figure 9 Bare cropland data for the Lampen Stream WFD waterbody (GB107040019790)

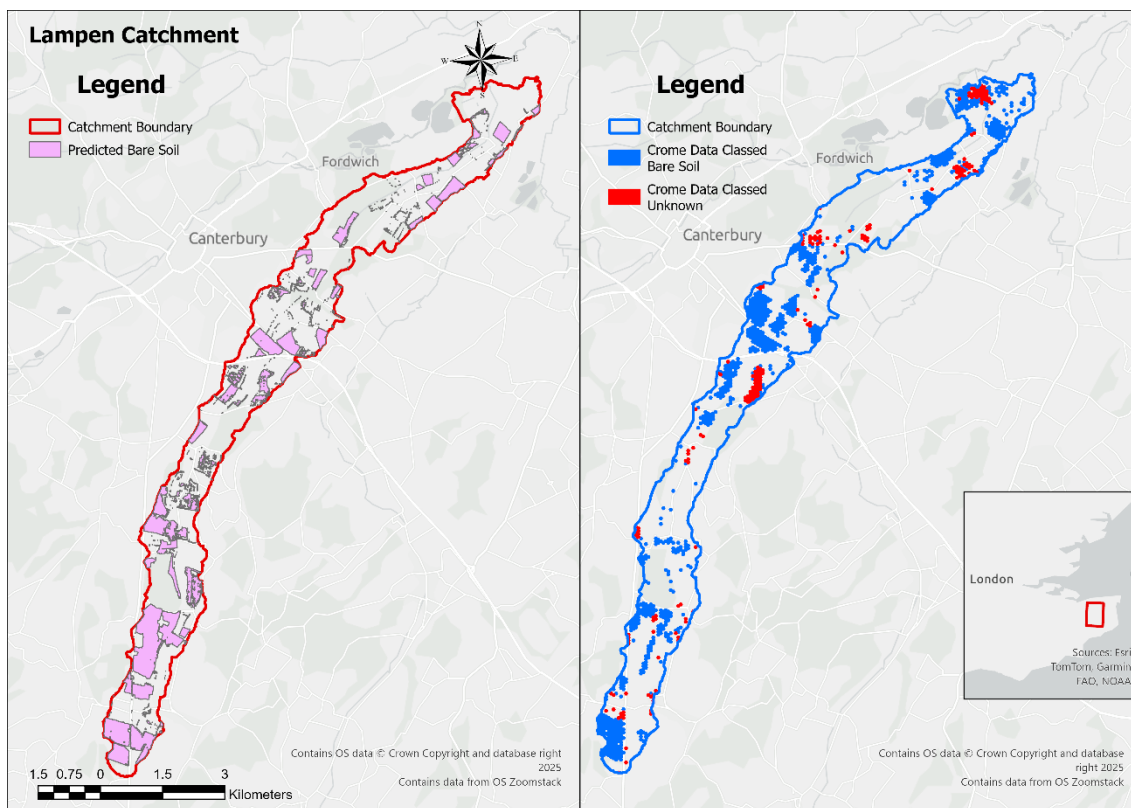


Figure 10 Bare cropland data for the Billingham WFD waterbody (GB103025072410)

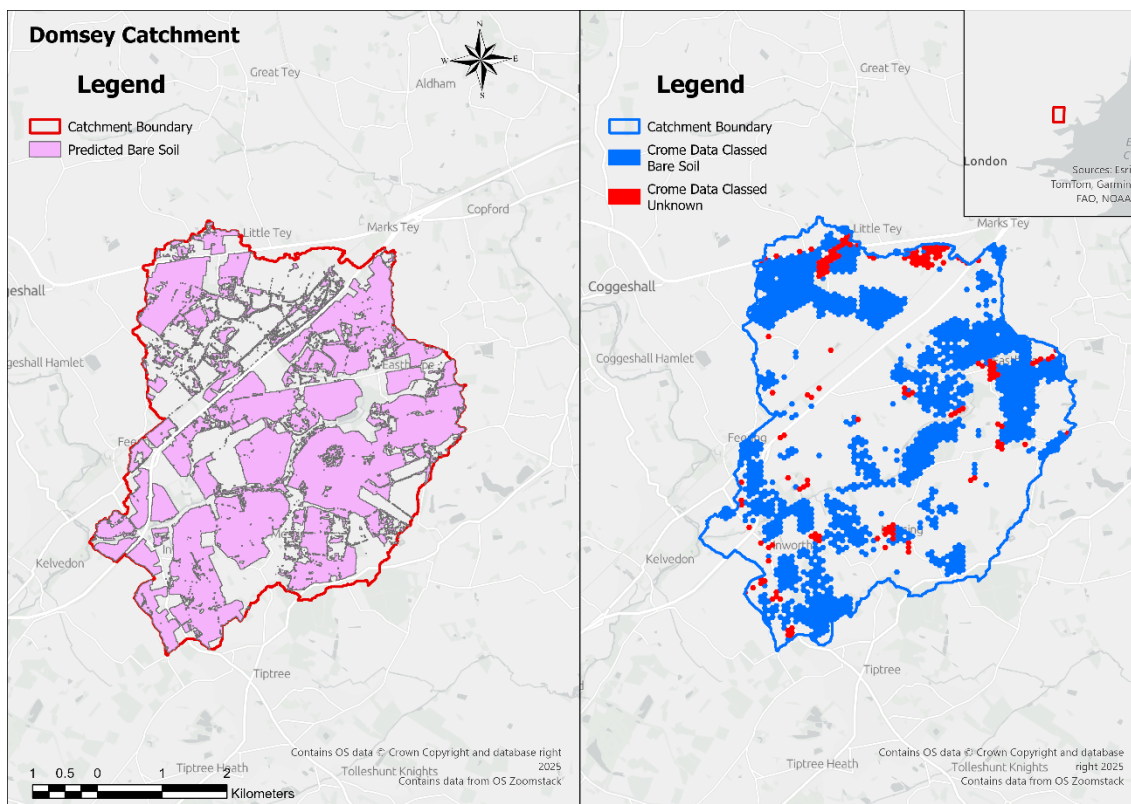


Figure 11 Bare cropland data for the Domsey Brook WFD waterbody (GB105037033870)

Table 2-3 Comparison of the predictions of bare cropland with the RDMAI model and inference from CROME data. Accuracy calculations assume that the CROME data is correct.

	RDMAI BARE		RDMAI CROP		Accuracy
	CROME Bare	CROME Crop	CROME Bare	CROME Crop	
BILLINGHAM	13	23	6	58	0.70
LAMPENSTREAM	9	14	8	69	0.78
BLACKFOSS	17	19	13	52	0.68
FILLINGHAM	23	27	13	36	0.60
DOMSEY	22	33	7	38	0.60
CHOLSEY	11	18	9	62	0.72
MOULTON	34	20	21	26	0.59
PLATT	9	21	17	54	0.63
SEELEY	14	17	5	65	0.78
GARREN	9	15	16	60	0.69

3. Benchmarking Conclusion

3.1 Summary

A summary table of key model characteristics can be found in Table 1-1, at the start of the report.

3.2 Conclusion

The Bare Cropland model was found to be fairly robust, with no obvious suggestions for how it could be improved. The time to run the model was relatively minor, with the benchmarking task taking a similar amount of time for both the RDMAI model and producing outputs via the CROME dataset, when factoring in obtaining required datasets and any preprocessing required.

Whilst the accuracy (or agreement) between the two approaches was only approximately 70%, the CROME land use data is not necessarily correct (with a stated classification accuracy based on ground truthing of c. 90%) and inferring over-winter crop cover based purely on land use will lead to some miss-classification (for example, due to the use of cover crops before spring sown crops). Thus, the true accuracy of the RDMAI model is likely to be higher. The model has the additional advantage that it is able to predict the duration of bare cropland, and should capture issues associated with poor crop establishment or performance.

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This report was prepared by Water Research Centre Limited (“WRc”) using the River Deep Mountain AI (“RDMAI”) model and such other models, benchmarking methodologies, data, documentation and information available to WRc at the time this report was prepared (together, the “Data Sources”).

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